Andrea Gaetani

Ph.D. Researcher in ICT



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Italy

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Birthday

31.12.1998

Driving license

B

Nationality

Italian

Links

Personal Portfolio
Linkedin
Github
Hack The Box

Languages

Italian - Native

English - B2

Work experience

Ph.D. Researcher in ICT, University of L'Aquila

October 2026 - Today / L'Aquila, Italy

Researching and designing secure AI solutions using local LLMs, ensuring that government data remains entirely private and compliant with strict data sovereignty regulations

AI Cloud Architect, Storm Reply S.r.l.

September 2025 - October 2026 / Rome, Italy

Designing and implementing cloud-native AI architectures on AWS to drive AIOps and end-to-end automation, speeding up workflows, strengthening observability, and improving scalability

AI Engineer Internship, Storm Reply S.r.l.

April 2025 - July 2025 / Rome, Italy

Implemented a multi-agent system for automated log analysis and reporting as part of a Master's thesis, enabling end-to-end ingestion, anomaly detection, root-cause triage, and executive reporting

Education

Bachelor Degree in Information Engineering, Università degli studi dell'Aquila
(September 2019 - December 2023)
Master Degree in Computer Engineering, Università degli studi dell'Aquila
(September 2023 - October 2025)

Skills

Ph.D. Research & Development Activities

- **Researching, designing, and deploying secure AI solutions and local LLM workflows for Public Administration**, optimizing open-source foundational models for on-premise infrastructure to guarantee absolute data sovereignty, users privacy, and compliance with institutional security standards

Cloud & Architecture

- AWS Solution Architecture, Serverless Architectures, Event-Driven Architectures, Workflow Orchestration & Automation, Scalability & Reliability Design (fault tolerance, decoupling, resiliency patterns), API-Driven System Design

AWS (Hands-on)

- AWS Lambda, Amazon S3, Amazon API Gateway, AWS Step Functions, Amazon ECR, Amazon CloudWatch, AWS CloudFormation, Amazon SageMaker, Amazon Bedrock, AgentCore, Amazon Connect

Observability / Operations

- Monitoring & alerting, dashboards, log-based troubleshooting, observability for distributed/serverless workloads

Automation & Delivery

- End-to-end process automation (low/no human intervention), workflow acceleration, operational efficiency improvements, CI/CD concepts, release automation

GenAI (Production)

- Amazon Bedrock, AgentCore (agent workflows, tool/function calling), LLM application architecture (prompt patterns, orchestration, integrations), Strands Agents (multi-agent workflows)

GenAI Frameworks (Prototyping & Evaluation)

- LangGraph (prototyping, production-readiness evaluation), CrewAI (prototyping, production-readiness evaluation), Agno (prototyping, production-readiness evaluation), Strands Agents (multi-agent production grade orchestration)